



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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CSA0208- C Programming

Integrated hospital management system with
patient records, doctor scheduling and billing

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INTRODUCTION

- An **Integrated Hospital Management System** is a computerized software platform developed to manage and coordinate different hospital activities through a centralized system.
- The system integrates important healthcare operations such as **patient registration, electronic medical records, doctor scheduling, appointment management, billing, and report generation.**
- Instead of maintaining separate paper-based records, the hospital can store important patient and administrative information in a centralized database.
- This system helps doctors, receptionists, administrators, and billing staff access the required information quickly and efficiently.
- By automating routine hospital operations, the system can reduce human errors, save time, and improve the overall quality of healthcare services.



PROBLEM STATEMENT



- Traditional hospital management systems often depend on **manual paperwork and separate record-keeping methods**, which can result in delays, data duplication, and difficulty in retrieving patient information.
- Managing doctor appointments manually can create **scheduling conflicts, long waiting times, and difficulties in identifying available doctors and consultation slots**.
- Maintaining billing information manually can lead to calculation errors, incorrect invoices, and difficulties in tracking payments and outstanding amounts.
- Patient medical records may become difficult to locate when they are maintained in physical files, especially when a patient has visited the hospital multiple times.
- Therefore, there is a need for an integrated digital system that can connect **patient records, doctor schedules, appointments, and billing information** in one centralized platform.



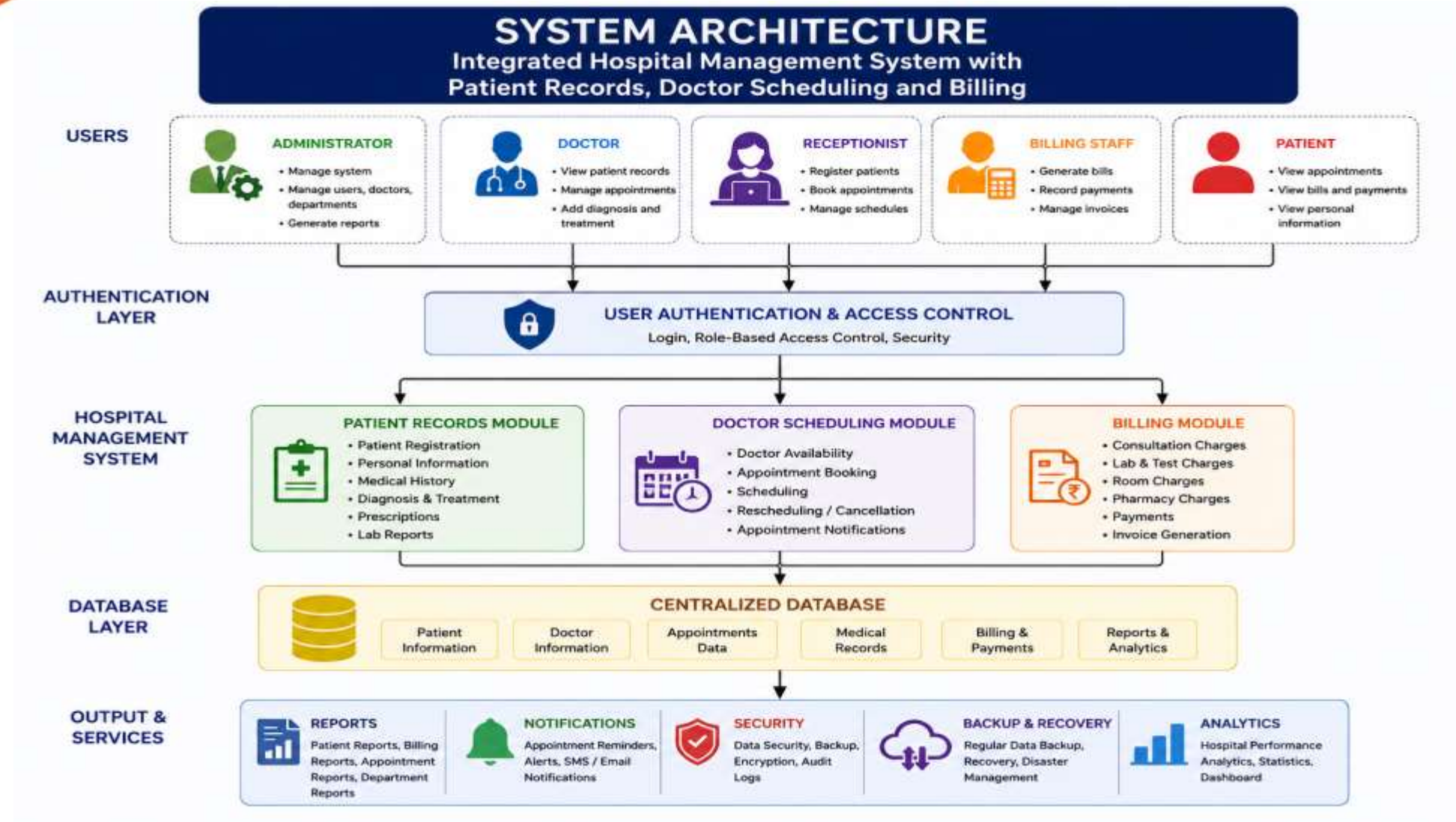
OBJECTIVE



- The primary objective of this project is to develop an **integrated hospital management system that simplifies and automates major healthcare and administrative operations.**
- The system aims to maintain complete and accurate patient records so that authorized doctors can quickly access medical history, diagnosis, prescriptions, and treatment information.
- Another important objective is to provide an efficient doctor scheduling system that allows hospital staff to manage doctor availability and patient appointments without scheduling conflicts.
- The billing module is designed to automatically calculate healthcare service charges and generate accurate invoices for patients.
- The system also aims to reduce paperwork, minimize human errors, improve data accessibility, and provide better service to patients and hospital staff.



SYSTEM ARCHITECTURE





Module 1: Patient Records Management



- The Patient Records Management module is responsible for maintaining **complete digital information about every patient registered in the hospital.**
- During registration, the system records patient details such as name, age, gender, contact information, address, and other necessary identification details.
- Doctors can update medical information including symptoms, diagnosis, treatment plans, prescriptions, laboratory results, and previous medical history.
- The system allows authorized healthcare professionals to quickly access previous patient records during consultations, which can support more informed treatment decisions.
- Digital storage also reduces the possibility of losing physical documents and makes patient information easier to retrieve whenever required.

```
--- Hospital Management System ---  
1. Add Patient  
2. View Patients  
3. Search Patient  
4. Update Patient  
5. Delete Patient  
6. Add Doctor  
7. View Doctors  
8. Schedule Appointment  
9. Generate Bill  
10. Exit  
Enter choice: 1  
Enter Patient ID: 01  
Enter Name: HARISHA  
Enter Age: 20  
Enter Disease: FEVER  
Patient added successfully!
```



Module 2: Doctor Scheduling and Appointment Management



- The Doctor Scheduling module is designed to **manage doctor availability and organize patient appointments in an efficient and systematic manner.**
- Hospital staff can enter the working hours, available consultation slots, specialization, and department of each doctor into the system.
- Patients or receptionists can select an appropriate doctor and available time slot based on the doctor's schedule.
- The system can prevent two patients from being assigned to the same doctor at the same time, thereby reducing scheduling conflicts.
- Automated appointment reminders can further help reduce missed appointments and improve patient convenience.
- Appointment cancellation and rescheduling can also be handled through the system, allowing hospital staff to maintain an updated schedule.

```
--- Hospital Management System ---
1. Add Patient
2. View Patients
3. Search Patient
4. Update Patient
5. Delete Patient
6. Add Doctor
7. View Doctors
8. Schedule Appointment
9. Generate Bill
10. Exit
Enter choice: 06
Enter Doctor ID: 1
Enter Name: USHA
Enter Specialization: ENT
Doctor added successfully!
```



Module 3: Billing Management



- The Billing Management module automates the process of **calculating hospital service charges and generating accurate bills for patients.**
- The system can include charges related to doctor consultation, laboratory tests, scanning services, medicines, room charges, procedures, and other medical services.
- When a patient receives different services, the corresponding charges can be automatically added to the patient's billing record.
- The system calculates the total amount and generates a detailed invoice containing the individual services, charges, discounts, taxes, and final payable amount.
- Payment information can also be recorded so that hospital staff can easily identify whether a bill is paid, partially paid, or still pending.
- This automated process reduces calculation mistakes and improves transparency in hospital financial management.

```
--- Hospital Management System ---
1. Add Patient
2. View Patients
3. Search Patient
4. Update Patient
5. Delete Patient
6. Add Doctor
7. View Doctors
8. Schedule Appointment
9. Generate Bill
10. Exit
Enter choice: 9
Enter Patient ID: 01
Consultation Fee: 200
Medicine Charge: 400
Bill generated. Total = 600.00
```



Sample Program Features

Main Functions

- Add Patient
- View Patient
- Search Patient
- Update Patient
- Delete Patient
- Add Doctor
- View Doctor
- Schedule Appointment
- Generate Bill
- Exit

Sample Data Structure

```
struct Patient
{
    int id;
    char name[30];
    int age;
    char disease[30];
};

struct Doctor
{
    int id;
    char name[30];
    char
specialization[30];
};

struct Bill
{
    int patientID;
    float
consultationFee;
    float
medicineCharge;
    float total;
};
```



ADVANTAGES AND APPLICATIONS



Advantages

- Easy patient record management
- Faster doctor scheduling
- Accurate billing
- Reduces paperwork
- Saves time
- User-friendly menu
- Secure data storage using file

Applications

- Hospitals
- Clinics
- Diagnostic Centers
- Health Care Centers
- Medical Colleges



FUTURE SCOPE



- In the future, the system can be enhanced by integrating **Artificial Intelligence and Machine Learning techniques** to support disease prediction, patient risk analysis, and intelligent healthcare recommendations.
- A mobile application can be developed so that patients can book appointments, view medical records, receive reminders, and check billing information using their smartphones.
- Online consultation and video conferencing features can be added to support remote healthcare services.
- Online payment gateways can be integrated to allow patients to securely pay hospital bills without visiting the billing counter.
- AI-powered chatbots can be introduced to answer common patient questions and provide basic hospital information.



CONCLUSION

- The **Integrated Hospital Management System** provides an efficient and centralized solution for managing important hospital activities such as patient records, doctor scheduling, appointments, and billing.
- By replacing many manual processes with automated digital workflows, the system can reduce paperwork, minimize errors, save time, and improve information accessibility.
- The integration of different modules allows hospital departments to share accurate and updated information while maintaining appropriate security and access control.
- The system can improve the overall experience of patients as well as the productivity of doctors, receptionists, administrators, and billing staff.
- With the future integration of **Artificial Intelligence, cloud computing, mobile applications, online payments, and telemedicine**, the proposed system can become a scalable and intelligent hospital management platform.



THANK YOU